

次作业

$$2 \quad \dot{U}_2 = \frac{\frac{R}{j\omega C}}{R + \frac{1}{j\omega C}} \quad \dot{U}_1 = \frac{\frac{R}{j\omega C}}{\frac{R}{j\omega C} + j\omega L + \frac{1}{j\omega C}} \quad \dot{U}_1 = \frac{R}{R - \omega^2 RLC + j\omega L} \dot{U}_1$$

$$= \frac{R^2 - \omega^2 R^2 LC^2 - j\omega L R}{(R - \omega^2 RLC)^2 + \omega^2 L^2} \dot{U}_1$$

$$H(j\omega) = \frac{\dot{U}_2}{\dot{U}_1} = \frac{R}{R - \omega^2 RLC + j\omega L}$$

ω 较大时, $H(j\omega) \rightarrow 0$ $\therefore$ 为低通特性

$$3 \quad \dot{U}_2 = \frac{\frac{R_2}{j\omega C_2}}{R_2 + \frac{1}{j\omega C_2}} \quad \dot{U}_1 = \frac{\frac{R_2 \cdot C_1}{C_2}}{R_2 + R_1 + j\omega C_1 \left(\frac{R_2 + \frac{1}{j\omega C_2}}{R_1 + \frac{1}{j\omega C_1}} \right)} \dot{U}_1$$

$$= \frac{\frac{R_2}{j\omega C_2}}{R_2 + \frac{1}{j\omega C_2}} + \frac{\frac{R_1}{j\omega C_1}}{R_1 + \frac{1}{j\omega C_1}}$$

$$\therefore R_1 C_1 = R_2 C_2 \quad \dot{U}_2 = \frac{j\omega R_2 C_1 R_1 + R_2}{- \omega C_1 C_2 R_2 + j\omega C_1} \dot{U}_1 = \frac{R_2}{j\omega C_1} \dot{U}_1$$

$$\therefore R_1 C_1 = R_2 C_2 \quad \dot{U}_2 = \frac{R_2}{j\omega C_1} \dot{U}_1$$

$$H(j\omega) = \frac{R_2}{j\omega C_1} \quad \text{故为低通特性}$$

4 谐振时, 有

$$X_C = X_L$$

$$\dot{I}_1 = \frac{\dot{U}_1}{R_1} = 0.5 A \quad \therefore \dot{I}_2 = 0.5 \dot{I}_S - \dot{I}_1 = 0.5 A$$

$$R_2 = \frac{\dot{U}_1}{\dot{I}_2} = 100 \Omega$$

$$\dot{U}_L = \dot{I}_2 \cdot jX_L = (50j) V$$

$$U_1 = 50 V$$

$$H(j\omega) = \frac{R_2}{R_2 + R_1}$$

5W 功率

$$\dot{U} = 0.1 \angle 0^\circ \text{ V}$$

4. 7.5 (1) 最大时即谐振时,

$$R = \frac{\dot{U}}{I_m} = 0.1 \Omega \quad \therefore U_L = 10 \text{ V} \quad U_L = 10 \angle 90^\circ \text{ V}$$

$$\therefore \frac{U_L}{I_m} = \frac{U_L}{I_m} = 10 \angle 90^\circ \Omega \quad \text{有 } \omega L = 10 \Omega \Rightarrow L = 10^{-3} \text{ H}$$

$$\therefore \text{谐振} \therefore \omega = \sqrt{\frac{1}{LC}} \Rightarrow C = 10^{-5} \text{ F}$$

$$Q = \frac{\omega L}{R} = 100$$

$$(2) \dot{U}_C = \frac{1}{j\omega C} \cdot \dot{I} = 10 \angle -90^\circ \text{ V} \quad \therefore u_C = 10\sqrt{2} \cos(\omega t - 90^\circ) \text{ V}$$

$$7.6 (1) \text{ 由 } \Delta\omega = \frac{\omega_0}{Q} \Rightarrow Q = 3.504$$

$$\therefore \omega_0 = \sqrt{\frac{1}{LC}} \Rightarrow C = \dots \therefore f_0 = \sqrt{\frac{1}{2\pi LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = 1.03 \times 10^{-7} \text{ F} \quad C = 1.03 \times 10^{-7} \text{ F}$$

$$\text{由 } Q = \frac{\omega L}{R} = \frac{2\pi f L}{R} \Rightarrow R = 5.03 \times 10^2 \Omega \quad R = 5.03 \times 10^2 \Omega$$

$$(2) f = 876 \text{ Hz 时 } I = \frac{U}{R} = 0.046 \text{ A}$$

$$P = UI = 1.07 \text{ W}$$

$$f = 750 \text{ Hz 时 } Z = R + j\omega L + \frac{1}{j\omega C} = 15032.552.29 \text{ j}$$

$$I = \frac{U}{Z} = (0.02 + 0.023 \text{ j}) \text{ A}$$

$$P = \operatorname{Re}(U \dot{I}^*) = 0.487 \text{ W}$$

$$f = 750 \text{ Hz} / 1000 \text{ Hz} \quad P' = \frac{P}{2} = 0.535 \text{ W}$$

~~$$f=1000\text{Hz时 } Z=R+j\omega L+\frac{1}{j\omega C}=(503+465.4j)\Omega$$~~

~~$$I=\frac{U}{Z}=(0.025-0.023j)\text{A}$$~~

~~$$P=\text{Re}(UI^*)=0.576\text{W}$$~~

13) 谐振时 $U_L = U_C = U_R \cdot Q = 81.3\text{V}$

7.9 顺接时, 等效电感为 (L_1+L_2+2M)

谐振角频率 $\omega_1 = \sqrt{\frac{1}{(L_1+L_2+2M)C}} = 1000\text{rad/s}$

谐振时 $I = \frac{U}{R_1+R_2} = 0.4\text{A}$

$\dot{U}_1 = j\omega(L_1+M)\dot{I} + \dot{I}R_1 = 8\angle 90^\circ\text{V} + 2\sqrt{2}\angle 45^\circ\text{V}$

$U_1 = 8\text{V} = U_1 = 8.25\text{V}$

$\dot{U}_2 = j\omega(L_2+M)\dot{I} + \dot{I}R_2 = 12\angle 90^\circ\text{V}$

$U_2 = 12\text{V} = U_2 = 12.65\text{V}$

反接时, 等效电感为 $(L_1+L_2-2M) = 450\angle 71.5^\circ\text{V}$

谐振角频率 $\omega_2 = \sqrt{\frac{1}{(L_1+L_2-2M)C}} = 2236\text{rad/s}$

谐振时 $I = \frac{U}{R_1+R_2} = 0.4\text{A}$

$\dot{U}_1 = j\omega(L_1-M)\dot{I} + \dot{I}R_1 = 2\text{V}$

$U_1 = 2\text{V}$

$\dot{U}_2 = j\omega(L_2-M)\dot{I} + \dot{I}R_2 = 9.80\angle 65.9^\circ\text{V}$

$U_2 = 9.80\text{V}$

7.11 $Z_{in} = R + \frac{1}{j\omega C_2 + \frac{1}{j\omega L + \frac{1}{j\omega C_1}}}$

$$= R + \frac{1}{j\omega C_2 + \frac{\omega C_1}{1 - \omega^2 L C_1}}$$

$$= R - \frac{1}{\omega C_2 + \frac{\omega C_1}{1 - \omega^2 L C_1}} = R - \frac{1}{0.5\Omega + \frac{1.5\Omega}{1 - 1.5\omega L}}$$

当 $|Z_{in}|$ 最小时, I 最大.

$$= R - \frac{1}{0.5 + \frac{1.5}{1 - \omega L}} = R - \frac{1.5\omega L}{1.5 - 0.5\omega L}$$

$$= R - \frac{2}{1.5 - 0.5\omega L} j$$

$$= R + (-2 + \frac{2}{1.5 - 0.5\omega L}) j$$

即当 $-2 + \frac{2}{1.5 - 0.5\omega L} = 0$ 时 即 $\omega L = 1\Omega$ 或 $L = 0.01H$ 时, 最大 $I_{max} = \frac{U}{R} = 2A$.

此时 $X_L + X_C = 0 \therefore \dot{U}_1 = I \cdot \frac{1}{j\omega C} \Rightarrow \dot{U}_1 = 2 \angle -90^\circ V$
 $\therefore u_1 = 2\sqrt{2} \cos(\omega t - 90^\circ) V$

(2) 当并联谐振时, I 最小.

(2) $\omega L = 3\Omega$
 即 $L = 0.03H$ 时

即有 $\omega L + \frac{1}{\omega C_1} = -\frac{1}{\omega C_2} \Rightarrow L = 0.03H$ (2分) $\rightarrow \infty$

等3分 $L = 0.02H$ $C = 0.5 \times 10^{-2} F$ $I = 0$

$R_0 = \frac{L}{RC} = 4\Omega$ $I_{min} = \frac{U}{R_0} = 0.5A$ $\dot{U}_L + \dot{U}_1 = 2 \angle 0^\circ V$

$\dot{U}_R = 0.5 \angle 0^\circ V$ $\frac{\dot{U}_L}{\dot{U}_1} = \frac{X_L}{X_{C1}} = -3 \Rightarrow \dot{U}_1 = -1 \angle 0^\circ V \Rightarrow u_1 = \sqrt{2} \cos \omega t$

$\therefore \dot{U}_L + \dot{U}_1 = 1.5 \angle 0^\circ V$ $\frac{\dot{U}_L}{\dot{U}_1} = \frac{X_L}{X_{C1}} = \frac{3j}{-j} = -3 \Rightarrow \dot{U}_1 = -1 \angle 0^\circ V \Rightarrow u_1 = \sqrt{2} \cos \omega t$

$\therefore \dot{U}_1 = 0.75 \angle 180^\circ V$ 即 $u_1 = 0.75\sqrt{2} \cos(\omega t + 180^\circ) V + 180^\circ$

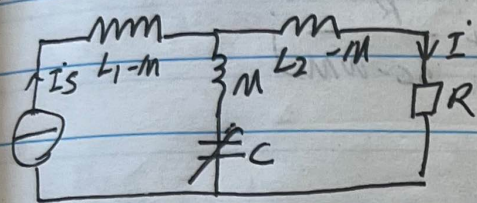
$$7.12 \quad Z_{eq} = j\omega L + \frac{(j\omega M)^2}{j\omega \cdot 2mH}$$

$$= j\omega \cdot 3mH - j\omega \frac{M^2}{2mH}$$

$$\therefore Z_{eq} = 3mH - \frac{M^2}{2mH} = \frac{5}{2}mH$$

$$\therefore \omega = \sqrt{\frac{1}{LC}} = 2 \times 10^5 \text{ rad/s}$$

7.13 等效电路如图T.



$$I = \frac{j\omega M + \frac{1}{j\omega C}}{j\omega M + \frac{1}{j\omega C} + j\omega(L_2 - M) + R} I_s$$

$$= \frac{-10^6 C + 1}{-3 \times 10^6 C + 1 + 12 \times 10^6 C j} I_s$$

$$= \frac{(-10^6 C + 1)(-3 \times 10^6 C + 1 - 2 \times 10^6 C j)}{(-3 \times 10^6 C + 1)^2 + 12 \times 10^6 C j^2} I_s$$

$$I = \frac{\omega^2 + \frac{1}{C}}{-3\omega^2 + \frac{1}{C} + j\omega R} I_s$$

$$I = \frac{3 + \frac{2}{C\omega^2} + \frac{j\omega R}{1 - \omega^2 C}}{C\omega^2 - 1} I_s$$

$$I = \frac{1 + \frac{2\omega^2 - j\omega R}{-3\omega^2 + \frac{1}{C} + j\omega R}}{C\omega^2 - 1} I_s$$

$$I = \frac{j\omega M + \frac{1}{j\omega C} - jX_C}{j\omega M + j\omega(L_2 - M) + \frac{1}{j\omega C} + R - jX_C} I_s$$

$$I_s = \frac{j\omega(L_2 - M)}{j\omega(L_2 - M) + R}$$

$$I_s = \frac{1}{\frac{L_2 - M}{M - X_C} + \frac{R}{\omega(M - C)j}}$$

$$= \frac{3 - X_C}{1 - X_C} - \left(\frac{2}{1 - X_C} \right) j$$

~~I 最小即当 $\left| \frac{3-c}{1-c} + \frac{2}{1-c}j \right|$ 最大~~

~~$$\begin{aligned} \therefore \sqrt{\frac{9-6c+c^2+4}{(1-c)^2}} &= \sqrt{\frac{c^2-6c+13}{c^2-2c+1}} = \sqrt{1 + \frac{12-4c}{c^2-2c+1}} = \sqrt{1 + \frac{4(3-c)}{(c-3)^2+4}} \\ &= \sqrt{1 - \frac{4}{(c-3)^2+4}} \end{aligned}$$~~

~~$\therefore$ 当 $c=3$ 时, 有最大值~~
~~当 $c=3=5/2$ 时~~

即 $c=$

$$\frac{1}{2} X_C = \omega \cdot t$$

$$\Rightarrow \dot{I} = \frac{1}{\frac{M-t}{L_2-t} + \frac{R}{\omega(t-M)}} = \frac{1}{\frac{3-t}{1-t} + \frac{2}{t-1}j}$$

分母复数模 = $\sqrt{\frac{t^2-6t+13}{(1-t)^2}} = \sqrt{1 - \frac{4}{(t-3)^2+4}}$

当 $t-3=2$ 时, 有最小值 $\sqrt{\frac{1}{2}}$ 此时 $I = (1-j)I_S$
~~当 $t=5$~~

$$\therefore I_{\max} = 10\sqrt{2} \text{ mA} \quad X_C = \frac{1}{\omega C} = 5 \Omega \Rightarrow C = 2 \times 10^{-7} \text{ F}$$

当 $t-3=0$ 时即 $t=3$ 时 $C = 3.33 \times 10^{-7} \text{ F}$ 时 分母复数模有最大值 ∞
 ~~1×10^{-6}~~

$$I_{\min} = 10 \text{ mA} \quad I_{\min} = 0 \text{ A}$$

在频率 ω 时

19 阻止基波通过负载，即 L_1, C 发生并联谐振

$$\text{有 } \omega = \frac{1}{2\pi f \sqrt{L_1 C}} \Rightarrow L_1 = 2.53 \times 10^{-4} \text{ H}$$

九次波顺利通过，即 $L_1 // C$ 与 L_2 发生串联谐振

电路阻抗为
$$j\omega L_2 + \frac{j\omega L_1 \cdot \frac{1}{j\omega C}}{j\omega L_1 - j \cdot \frac{1}{\omega C}}$$

$$= j\omega L_2 + \frac{\frac{L_1}{C}}{\omega L_1 - \frac{1}{\omega C}} (1-j)$$

$$\text{即有 } 9\omega L_2 - \frac{\frac{L_1}{C}}{9\omega L_1 - \frac{1}{9\omega C}} = 0$$

$$\omega = 2\pi f \Rightarrow L_2 = \cancel{3.86 \times 10^{-8} \text{ H}} = 3.16 \times 10^{-6} \text{ H}$$